

ARNAV NEGI

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Skills

- **Programming:** Python, SQL (MySQL, Oracle PL/SQL), Bash, C, C++, MATLAB
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, PyTorch, Clustering (DBSCAN), Model Evaluation, Ensemble Methods, Deep Learning
- **Computer Vision:** Semantic Segmentation, Segment Anything (SAM), SegFormer, DeepLabV3+, CLIP, OpenCV
- **Data Analysis:** Pandas, NumPy, EDA, Data Cleaning, Feature Engineering, Geospatial Analysis
- **NLP & LLMs:** Transformers, Hugging Face, LangChain, LangGraph, Vector Databases (Chroma, Milvus), Ollama, OpenAI API, Gemini API, RAG Systems
- **Cloud & Systems:** AWS, Docker, Cloud Computing
- **Web Development:** Django, Bootstrap
- **Visualization:** Power BI, Matplotlib, Streamlit, Folium
- **Tools:** Git, GitHub, Jupyter, Google Colab, VS Code
- **Soft Skills:** Leadership, Stakeholder Engagement, Public Speaking, Cross-cultural Communication

Education

University of Queensland

Jul 2024 – Jul 2026

Master of Computer Science (Management)

Brisbane, Australia

- Relevant Coursework: Machine Learning, Artificial Intelligence (Search & Planning), Natural Language Processing, Information Retrieval, Data Mining, Cloud Computing

Thapar Institute of Engineering and Technology

Aug 2020 – Jun 2024

Bachelor of Engineering in Computer Engineering

Punjab, India

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks

Professional Experience

APS (Association of Postgraduate Students)

Dec 2025 – Jul 2026

President (Coursework) - Part-time Leadership Role

Brisbane, Australia

- Represented 10,000+ coursework postgraduates on APS's Postgraduate Policy Board, helping shape positions on 2 major UQ policy changes (secure assessment thresholds, HDR unpaid work guidelines) by reviewing reports from representatives across 3+ university committees, including TLSEC.
- Helped allocate \$58,000 in bursaries supporting attendance at 80+ conferences in 2026 by assessing each applicant's evidence (registration, budget, academic relevance) as a member of the decision panel.
- Helped secure ~\$1M in funding for APS by representing postgraduate needs at Student Services and Amenities Fee (SSAF) general meetings.
- Initiated 2 new partnerships, with UQ's Student Enrichment and Success (SES) team and the Neurodivergent Hub, to broaden support for postgraduate students.
- Delivered 8–10 engagement events per semester reaching 500+ attendees by leading 15+ student representatives.

University of Queensland (QAAFI, Hayes Research Group)

Jan 2026 – Feb 2026

Data Science Intern, Genomic Prediction

Brisbane, Australia

- Analysed 50+ research papers on machine learning and deep learning for genomic prediction, summarising key knowledge gaps in a 7-page mini-review and presenting the findings to a Research Fellow to support their ongoing research.
- Benchmarked 6 genomic prediction models in Python on a wheat dataset, showing machine learning performed on par with the field-standard statistical approach.

Integra Micro Systems

Jun 2023 – Aug 2023

Software Development Intern, botAIML (AI/ML Division)

Bengaluru, India

- Built LangChain-based conversational chatbots as part of Integra's botAIML team, speeding up information retrieval by ~40% across a 10,000+ document knowledge base.
- Designed a hybrid keyword and semantic search pipeline, improving answer relevance for domain-specific queries.
- Reduced chatbot response latency by ~25% by refining prompts and streamlining the response generation workflow.

Technical Projects

Dwell-Aware TravelPlanner - Temporal Feasibility for LLM Itineraries | GitHub

- Cut overloaded itinerary days from 15.5% to 2.5% and raised full-plan feasibility from 48% to 90% by adding predicted visit durations and a daily time-budget check to the OSU-NLP TravelPlanner LLM agent.
- Built visit-duration labels for 3,346 places by clustering 1.3M+ check-ins with DBSCAN into 117k+ clean visit episodes, enriched with Google Places and OpenStreetMap features.
- Trained 5 regression models to predict how long people stay at each place, selecting Random Forest (22-minute test error), and compared against a reproduced GPT-3.5 baseline on 100 benchmark queries.

CoralReef AI - Multimodal Agentic RAG Chatbot for Coral Reef Education | GitHub

- Built a multimodal agentic RAG chatbot for coral reef education that scored best overall against 4 baselines and 2 ablations on a 35-query benchmark, with 94.9 tool-selection F1 and 94.3% image retrieval hit rate.
- Designed an intent gateway in LangGraph that plans which tools each question needs (text retrieval, image search, vision, memory) before running them, followed by a verifier step that catches unsupported claims.
- Indexed 13,000+ text chunks and coral images in Chroma, using CLIP for image search and Gemini for uploaded photos.

StreetView Urban Features - Semantic Segmentation for Urban Spatial Analysis | GitHub

- Measured 5 urban form indicators, including greenery (Green View Index, 21.7% average), across 206 Brisbane street-view images by building a Mapillary-to-segmentation pipeline sampled on a 15×15 grid.
- Benchmarked DeepLabV3+ on Cityscapes (76.9% mIoU, 96.0% pixel accuracy), then switched to SegFormer after finding that DeepLabV3+ handled Brisbane scenes poorly.

SAM Prompt Sensitivity - Prompt-Conditioned Image Segmentation Analysis | GitHub

- Tested how Meta's Segment Anything Model (ViT-B) responds to point, box and automatic prompts across 8 varied scenes, finding that masks agreed at only 0.46–0.64 mean IoU depending on the prompt pair.
- Showed that prompt sensitivity depends on the scene, with agreement reaching 0.99 IoU on isolated objects but falling to 0.13 in crowded scenes, using a reproducible PyTorch and OpenCV pipeline on GPU.

ResearchDoc - AI-Powered Research Management Platform

- Built a full-stack research management platform with Django and MySQL, supporting project workflows, resource management and subscription archiving.
- Integrated the OpenAI API so users can generate, review and save structured summaries of their research resources.

Leadership, Activities & Engagement

UQ Business School Postgraduate Association

Feb 2025 – Dec 2025

Director of Corporate Relationships (Executive)

Brisbane, Australia

- Organised 4+ networking and corporate events reaching 400+ total attendees (100–150 per event), connecting postgraduate students with industry professionals.
- Managed 5+ corporate sponsor and partner relationships, coordinating event delivery, marketing and outreach.

APS (Association of Postgraduate Students)

Jul 2025 – Dec 2025

International Officer

Brisbane, Australia

- Increased cultural event attendance by ~30% over the previous semester by building and running a calendar of 6+ major cultural festivals for a 5,000+ international student community.
- Coordinated a 10+ member student team to deliver the festival program.

University of Queensland

Nov 2025

Leaders at EAIT Pathway Program

Brisbane, Australia

- Selected for this competitive program, completing 20+ hours of training in self-leadership, mentoring and diversity.

Additional Achievements & Activities

University of Queensland

Jul 2024 – Jul 2025

Campus Kitchen Volunteer

Brisbane, Australia

- Supported meal preparation and service for 100+ students per session in a fast-paced team environment.

UQ Union

Jul 2024 – Jul 2025

Cloth Shop Volunteer Assistant

Brisbane, Australia

- Assisted with retail operations including inventory management, customer service and store organisation.

- Completed a 12-day high-altitude cycling expedition from Manali to Ladakh, summiting Khardung La (18,380 ft) and Tanglang La (17,580 ft), among the world's highest motorable passes.
- Won the Inter-State Padmashree Declamation Contest in Shimla.

Referees

Dr Seema Yadav

Research Fellow, Predictive Agriculture

Centre for Animal Science, QAAFI

The University of Queensland

Relationship: Internship Supervisor

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Richard Lee

Board Chair

APS (Association of Postgraduate Students)

The University of Queensland

Relationship: Worked with during my APS Presidency

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